CHAPTER 6

HUMAN IDENTIFICATION USING MMWAVE RADAR

This chapter introduces an innovative approach using deep learning for human identification utilizing mmWave radar technology. Unlike conventional vision methods, the approach in this work ensures privacy and accuracy in various indoor settings. Leveraging partial PointNet, MLP, CNN, and Bi-LSTM network components, this chapter proposes a unique neural network structure named MMIDNet, designed to process point cloud data from mmWave radar directly. The system achieves an impressive identification accuracy of 92.4% for 12 individuals. The research encompasses initial evaluation, data collection, system design, and system evaluation, highlighting the potential of mmWave radar combined with deep learning for secure and efficient human identification in IoT applications. The content of this chapter has been partially published in the MECO conference 2024 [53][1].

The sections are organized as follows: Section 6.1 provides a brief overview of this chapter. Section 6.2 evaluates the human identification task using the FAUST dataset, demonstrating the feasibility of using deep learning techniques to identify human targets by using point clouds. Following this, Section 6.3 details the preparation and pre-processing of the point cloud dataset collected using mmWave radar. Section 6.4 presents the specifics of the network structure and explains its contribution to the human identification problem. Section 6.5 provides a comprehensive evaluation of the system, covering overall performance as well as the effects of different network configurations, observation durations, and numbers of people. It also includes a comparison with other researchers' works. Finally, Section 6.6 provides a conclusion for this chapter.

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6.1 Overview

In the evolving landscape of sensing technologies, the precise identification of human targets within indoor environments stands as a critical endeavour with profound implications for various applications. Whether in the realms of in-house security, healthcare, or smart building management, the accurate and unobtrusive identification of individuals within indoor spaces is essential. Traditional sensing methods, including optical cameras [4, 25, 67] and infrared sensors [118], have been foundational in this pursuit. However, these methods frequently encounter challenges like limited visibility, difficulties in adapting to varying lighting conditions, and privacy concerns.

To overcome these challenges, the emergence of mmWave radar technology offers an innovative solution for indoor human target identification. Many researchers explored applications of human identification, aiming to distinguish users from each other and detect intruders from users using point clouds obtained by mmWave radar [27, 48, 49, 127, 192].

In this study, three FMCW radar IWR1843 development boards from TI are utilized to generate point clouds within the indoor environments for the training, validation, and test datasets. This chapter explores the potential of harnessing mmWave radar technology for human identification and proposes a deep neural network tailored for datasets in point cloud format to enhance accuracy. The primary contributions of this work include:

- The feasibility of employing deep learning techniques for identifying human targets through point clouds is assessed by using the FAUST dataset.
- A neural network structure, named MMIDNet, is designed to directly process point clouds from mmWave radars.
- An overall evaluation of the system is performed, demonstrating a notable identification accuracy of 92.4% for 12 individuals. The evaluation also explores the effects of various network configurations, observation durations, and the number of people on system performance.
- The work is available online at `https://github.com/DarkSZChao/MMWave_Radar_Human_ID_Classification` (accessed on 2 June 2024) to further promote the work in this field.

6.2 FAUST Evaluation

Human identification typically requires a significant amount of information, such as facial features, and the point cloud data from mmWave radar may be limited due to its sparse nature. Therefore, before using point clouds collected from mmWave radars for human identification, it's crucial to validate the feasibility of this approach. To accomplish this, an initial validation step is conducted using the FAUST dataset. This dataset comprises human models represented as watertight triangulated meshes, derived from a sophisticated high-resolution camera system

incorporating stereo cameras, RGB cameras, and speckle projectors [201]. Within the FAUST dataset, there are 10 subjects, each associated with 30 static postures, of which 10 postures are accompanied by aligned watertight models, resulting in a total of 100 models that can be used in this evaluation. Figure 6.1 shows some examples of the mesh models in FAUST with sampled point clouds. If accurate classification of these 10 identities can be achieved using the sparse point clouds, then the feasibility of this approach for human identification can be verified.

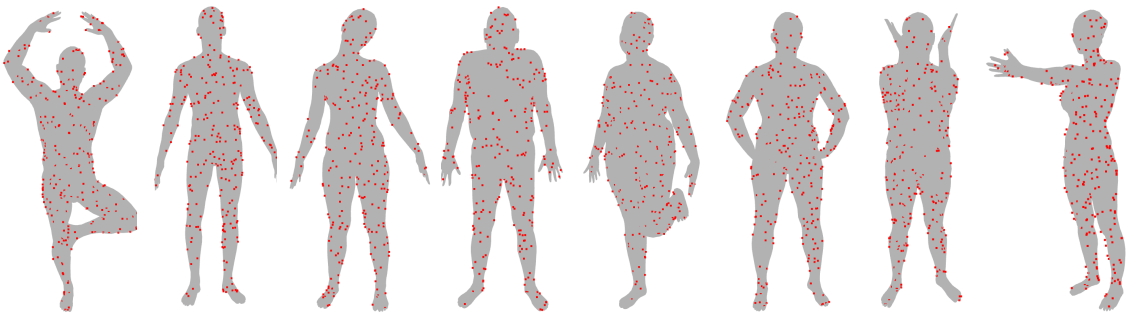


Figure 6.1: The example images for FAUST mesh models and point clouds. Red dots are points sampled on the mesh models.

6.2.1 Point Cloud Generation

Since the point clouds obtained from the mmWave radar comprise points defined by x, y, z coordinates, as elaborated in Section 5.3.2, the consistency must be maintained by extracting points with similar properties (e.g., point density) from the FAUST mesh models. Because the number of points received by mmWave radars in the experiment typically falls below 200 points per frame, randomly selecting 100 to 200 points from the surface of each mesh model as 1 sample simulates this scenario, with 100 samples taken from each model. However, neural networks struggle with data of varying lengths. Therefore, all samples are padded with zeros to maintain a consistent length of 200. The resulting dataset is structured as $N \times P \times C$ as shown in Equation (6.1), where N denotes the number of samples ($N = 10000$), P signifies the number of points in each sample ($P = 200$), and C represents the number of feature channels ($C = 3$, i.e., x, y, z coordinates). The elements of this tensor can be denoted as $X_{n,p,c}$, with n, p, c corresponding to the respective indices.

$$X \in \mathbb{R}^{N \times P \times C} \tag{6.1}$$

6.2.1.1 Data Augmentation

To address the issue of insufficient data, a coordinate field is created to simulate scenarios where subjects are present in a room. Random positional offsets are introduced to the point cloud

samples, ranging from 0 to 3 meters along the x and y axes (excluding the z-axis). Additionally, a random rotation, ranging from 0 to 360 degrees, is applied along the z-axis. To preserve the relative positions between points, which represent body shape, all points within each sample undergo the same rigid transformations for both translation and rotation.

Finally, to partition the dataset of 10,000 samples from 10 identities equally, 1,000 samples per identity are divided into training (70%), validation (10%), and testing (20%) sets. Each identity maintains the same proportion across these sets. The dataset is then shuffled in preparation for neural network training.

6.2.2 Network Design

MLP and CNN are two widely used neural network structures for spatial tasks such as image processing. They are able to capture relationships between data (pixels) efficiently to extract features. This section presents these two structures to perform the FAUST dataset evaluation. The fundamental details for MLP and CNN structures can be found in the previous Section 3.2. The networks in this section are deployed by using the Keras (version 2.6.0) deep learning framework in Python (version 3.8.6).

6.2.2.1 MLP

In this setup, two configurations of the MLP structure are prepared to evaluate the FAUST dataset comprehensively. One configuration uses five hidden dense layers with 3593k parameters, while the second uses three hidden dense layers with 1100k parameters. Figure 6.2 illustrates the configuration with five hidden dense layers. Given the input layer of size $N \times 200 \times 3$ with

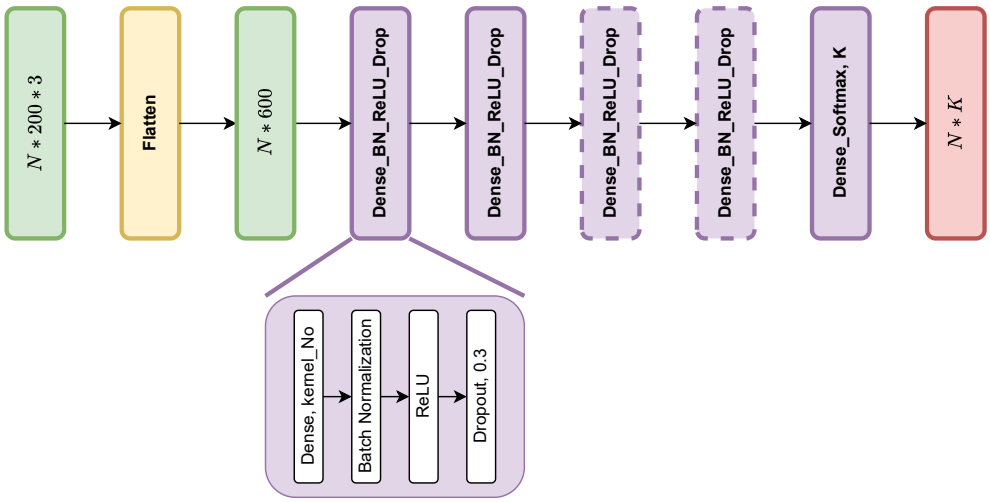


Figure 6.2: The MLP network structure with N samples and K categories. The dotted boxes represent the layers removed for various parameters in evaluation.

three dimensions, shown in green in Figure 6.2, a flatten layer is applied to convert the input to $N \times 600$ before passing it through the dense layers. Then, four dense layers are interconnected with batch normalization layers, ReLU activation layers, and dropout layers. The final dense layer, with a softmax activation function, outputs K people categories ($K = 10$ for the FAUST dataset). The configuration with three hidden dense layers follows a similar structure but omits the two dense layers, as drawn in dotted boxes in Figure 6.2, and has fewer total parameters.

6.2.2.2 CNN + MLP

In classification tasks, CNN networks are typically combined with MLP networks at the end to output the correct number of categories. Following the same strategy in the last section, three configurations that utilize both CNN and MLP are provided. As shown in Figure 6.3, the first configuration includes five convolutional layers with 2274k parameters, each convolutional layer specified with a particular number of kernels and kernel size. These convolutional layers are followed by max pooling layers, which reduce the data scale by combining the outputs of neuron clusters at the current layer into a single neuron in the next layer. A flatten layer is placed after the convolutional layers to stretch the data and reduce the data dimensions. Finally, a partial MLP network is applied after the CNN to perform classification into K categories. The second configuration uses three convolutional layers with 824k parameters, while the third configuration uses only one convolutional layer with 216k parameters. The omitted layers are indicated by dotted boxes in Figure 6.3.

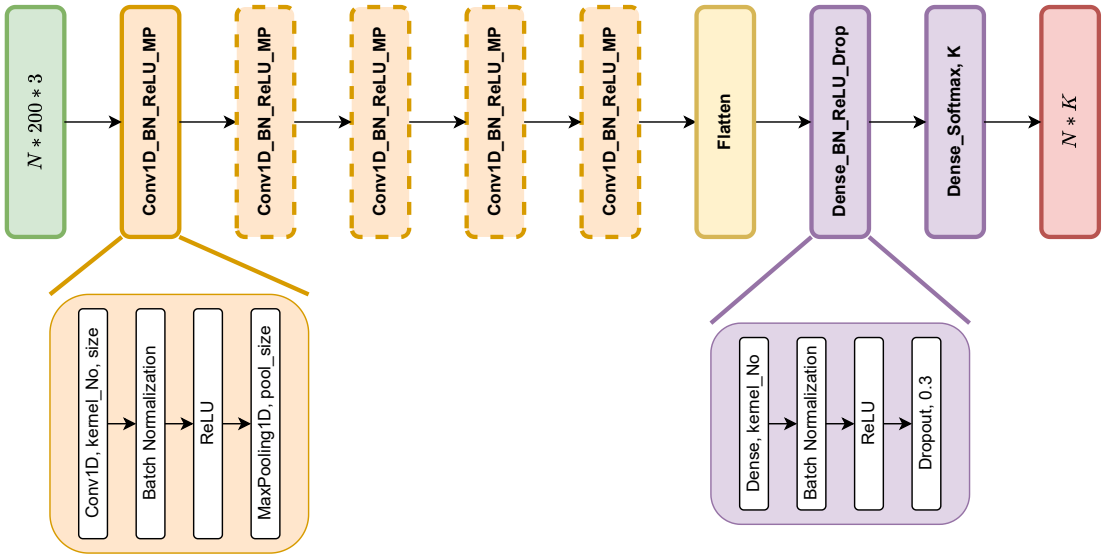


Figure 6.3: The network structure for CNN and MLP with N samples and K categories. The dotted boxes represent the layers removed for various parameters in evaluation.

6.2.3 Results and Comparison

Cross-entropy loss is employed as the loss function to quantify the difference between predicted and true probability distributions in classification tasks. Meanwhile, Adam is utilized as the optimizer, dynamically adjusting learning rates to ensure faster convergence for the networks. Training proceeds for 300 epochs, employing a learning rate of 0.0002 and a batch size of 128. Regarding the MLP network structure, validation accuracy on FAUST (shown in Figure 6.4 (a)) demonstrates that a single MLP network reaches a performance limit of around 40% for both parameter configurations of 1100k (three dense layers) and 3593k (five dense layers). Additionally, (b) depicts the loss for MLP on the validation set, which initially decreases until around 160 epochs before rising in the last 140 epochs. That indicates overfitting tendencies starting from 160 epochs. Overall, the single MLP structure exhibits poor performance in classifying human identification on the FAUST dataset.

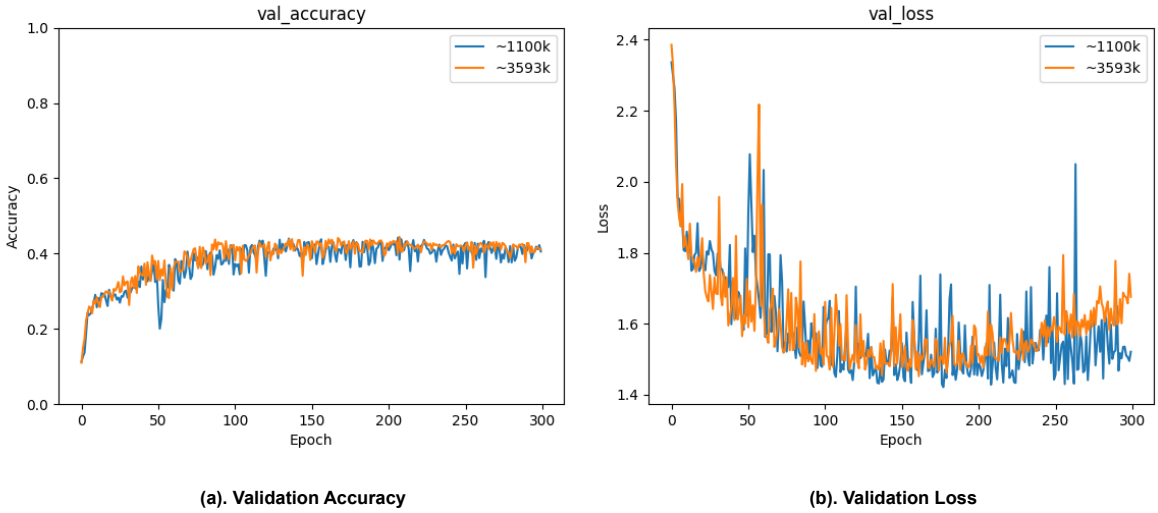


Figure 6.4: The accuracy (a) and loss (b) for a single MLP network on the validation set over epochs.

For the structure of CNN + MLP, there is a significant improvement in performance from 40% to 70% above, compared to the single MLP. As shown in Figure 6.5, on the validation set, networks with parameters of 216k (one convolutional layer) and 824k (three convolutional layers) achieve around 70% accuracy (a) with a loss value approaching 1.4 (b). Moreover, by expanding the network layers to five and increasing the parameter number to around 2274k, the network model can reach 80% accuracy and maintain stability with a lower loss below 1. This model also demonstrates a high accuracy level of 80% for 10 categories on the test set.

In conclusion, the CNN + MLP network structure outperforms the single MLP in the human classification task on the FAUST dataset. The achieved accuracy level of 80% proves the feasibility of applying neural networks to such classification tasks with the input of point clouds. This success

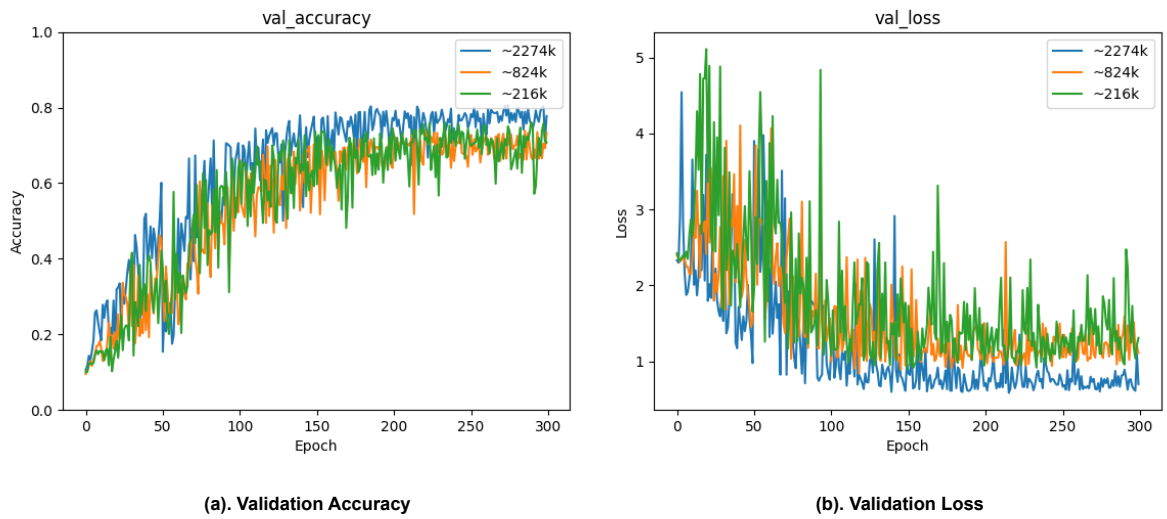


Figure 6.5: The accuracy (a) and loss (b) for CNN + MLP network on the validation set over epochs.

motivates further exploration into designing a network structure that can effectively handle point clouds from mmWave radars for human identification and classification.

6.3 Data Preparation

6.3.1 Data Collection

The data collection process follows the experimental setup described in Section 5.3. Point clouds are gathered from 12 randomly selected individuals, each walking around the radar room for 1-2 minutes per round, with a minimum of 10 rounds, resulting in approximately 20 minutes of data per participant. The subjects, aged 20-30, exhibit a range of heights (160-185 cm) and weights (50-80 kg), representing both male and female participants. The video footage is only utilized to ensure accurate labelling of the data and is not intended for experimental use. This data collection received approval from the Faculty of Engineering Research Ethics Committee at the University of Bristol (ethics approval reference code 17605).

6.3.2 Data Preprocessing

To maintain consistency, all the cloud points are expressed in the coordinate format (x, y, z, v, snr), where (x, y, z) denotes the point coordinates, and (v, snr) represents velocity and SNR level, respectively. A global coordinate system is established, facilitating the mapping of points collected from different radars through spatial rotation and position shift. Following the concatenation of data points from three radars, the points undergo filtering based on their positions and SNR

levels. Subsequently, the dynamic DBSCAN clustering algorithm from the previous Chapter 5 is applied, as shown in Figure 6.6.

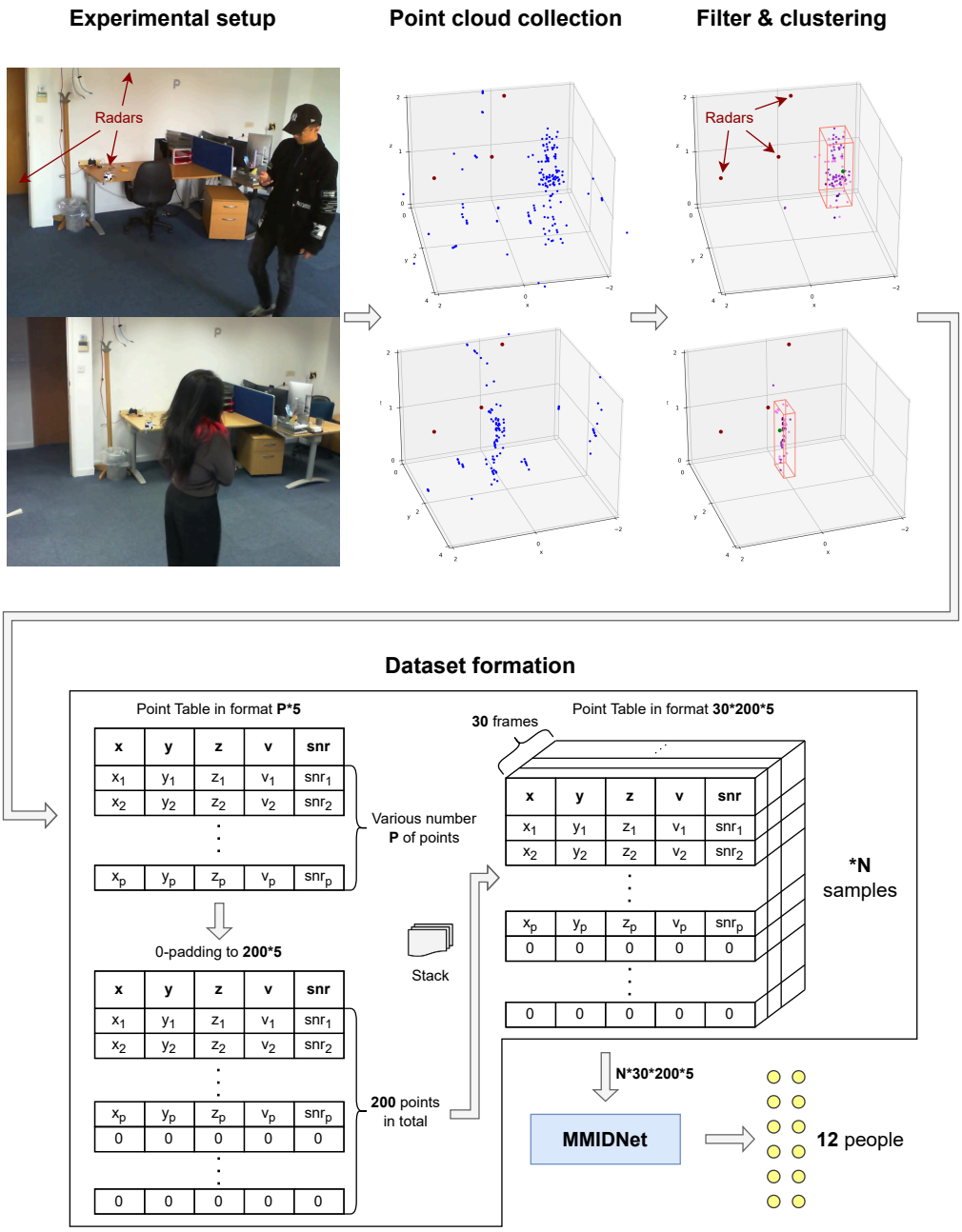


Figure 6.6: The system flowchart encompasses the experimental setup, point cloud collection, data filtering & clustering, and dataset preparation for network training.

In contrast to the approach outlined in [49], the network dataset in this work does not employ voxelization. Voxelization, as described in that work, incurs a high storage cost due to sparse voxelized data, posing challenges for GPU memory during accelerated training. Instead, a more

space-efficient point table representation is utilized. For instance, representing a human body in [49] requires a voxelized data size of $20 \times 20 \times 40$ to capture point cloud positions only for a single frame. In the method used here, the data is represented in a shape of $P \times 5$ (where P is the number of points in a given frame and $P \leq 200$) to represent a body, in terms of coordinates, velocity and SNR levels for the point cloud (depicted in Figure 6.6).

For each frame, the number of points (P) clustered by the DBSCAN algorithm exhibits variability due to the point cloud characteristics, resulting in a shape of $P \times 5$. To enable the network to analyze temporal data, all frames share a consistent shape of 200×5 by using zero-padding. Upon statistical analysis of the data, it is determined that the length of 200 points is optimal. Then, window sliding (length of 30 frames, step of 5 frames, overlap rate of 83%) is used to concatenate 30 neighbouring frames (approximately 1.5 seconds) to form a data sample. Following concatenation, one data sample transforms into the size of $30 \times 200 \times 5$.

The resulting dataset is organized as $N \times T \times P \times C$ as shown in Equation (6.2), where N is the number of samples, T represents the number of frames across time ($T = 30$), P indicates the number of points in each frame ($P = 200$) and C denotes the number of feature channels ($C = 5$). The elements of this tensor can be denoted as $X_{n,t,p,c}$, where n, t, p, c represents corresponding indexes.

$$X \in \mathbb{R}^{N \times T \times P \times C} \tag{6.2}$$

This structured dataset is then utilized to feed the MMIDNet network, which is proficient in classifying the identities of 12 individuals. The data from the 12 individuals used for identity classification is partitioned into 70% for training the network, 10% for validation, and 20% for testing. To enhance speed and reduce GPU memory usage during training, all the data is quantized to float16 data type.

6.4 System Design

6.4.1 Network Structure

MMIDNet is introduced as a neural network structure meticulously tailored and optimized for processing point clouds with five distinct features (x, y, z, v, snr) obtained from mmWave radars. The network comprises five key blocks: the Transformation Block, Residual CNN Block, Global Max Pooling, Bi-LSTM Block, and Dense Block, as illustrated in Figure 6.7.

6.4.1.1 Transformation Block

As mentioned in Section 3.2.4.1, in scenarios where a human body is represented using a 3D point cloud, applying rigid transformations such as translating or rotating on all points together does not lead to information loss or significant alterations, except for changes in the body's relative

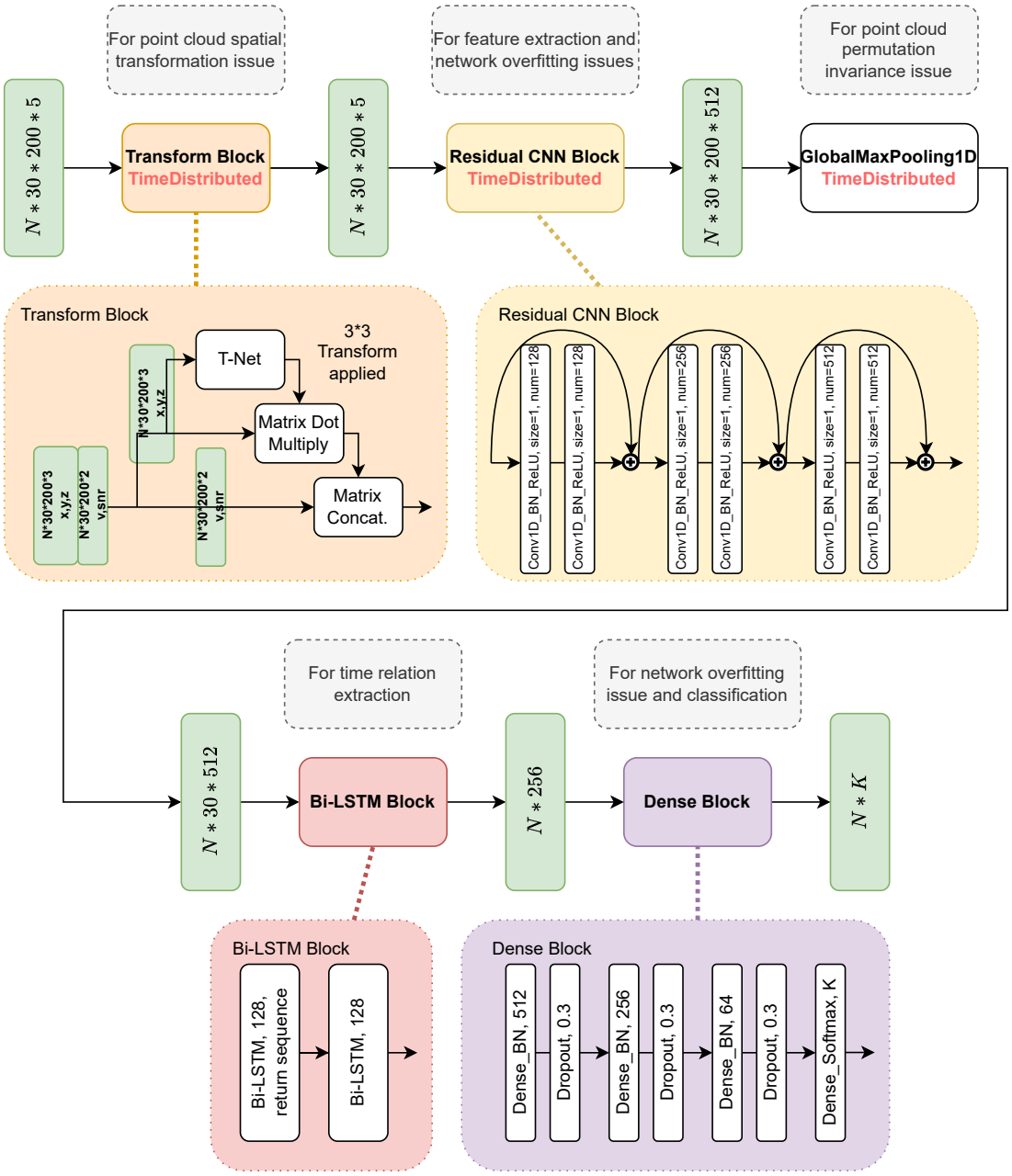


Figure 6.7: The proposed MMIDNet structure for human identify classification. K represents the number of identities ($K = 12$).

position [128]. Consequently, the system should maintain robustness on learned features when processing transformed 3D point cloud inputs.

Inspired by the usage of T-Net in PointNet [128, 202] (mentioned in Section 3.2.4), the data points (X_{input}) are initially separated on feature channels C dimension. The first coordinate segment has three features $C_{\{0,1,2\}}$ (x, y, z) organized as $N \times T \times P \times 3$, while the other two features

$C_{\{3,4\}}$ (v, snr) form a characteristic segment organized as $N \times T \times P \times 2$, as shown in Figure 6.7 and Equation (6.3). After that, the T-Net method capable of predicting an affine transformation matrix with a shape of 3×3 is adopted. This matrix is directly applied to the x, y, and z channels of the points by matrix dot multiplication. Finally, the previously isolated v and snr channels are concatenated back together. This series of actions constitutes the Transform Block.

$$X_{\text{input}} = \begin{bmatrix} X_{:,:,:,0:3} & \text{(First 3 channels in } C) \\ X_{:,:,:,3:5} & \text{(Last 2 channels in } C) \end{bmatrix} \tag{6.3}$$

6.4.1.2 Residual CNN Block

Within this block, a residual CNN structure is employed based on the residual structure [161]. Multiple CNN layers are utilized to extract features from the input points. To prevent potential degradation issues caused by stacking too many CNN layers, two CNN layers with identical hyperparameters are grouped into a residual block. The input and output of this group are connected by matrix addition, creating the input for the subsequent residual group, as illustrated in Figure 6.7. This approach helps prevent degradation issues in the network. Furthermore, a batch normalization layer and ReLu activation function following each CNN layer can standardize the input features to accelerate network learning convergence and avoid the overfitting issue [177].

6.4.1.3 Global Max Pooling

As discussed in Section 3.2.4.2, a point cloud is an unordered collection of vectors, which sets it apart from the structured pixel arrays in images or voxel arrays in volumetric grids, as used in [48, 49]. Specifically, as illustrated in Figure 6.8, two identical point cloud sets have the same geometric distribution but can be represented in different orders in table representation. Consequently, when a network processes 3D point clouds, it must preserve invariance to the permutation of the input set's data feeding order. In other words, the system should treat these differently ordered table inputs as equivalent since they represent the same point cloud.

To address this permutation invariance issue, based on [128, 202], a simple symmetric function called global max pooling layer is applied after the CNN in Figure 6.7. This pooling layer takes the maximum value across all dimensions (channels) for each feature, outputting a feature vector that remains invariant to the input vectors' order. This is because the maximum value for each feature is calculated independently of the order in which the points are arranged, ensuring a consistent representation regardless of the point cloud's permutation. The effectiveness of this approach has been demonstrated in point cloud applications [128], showcasing strong performance.

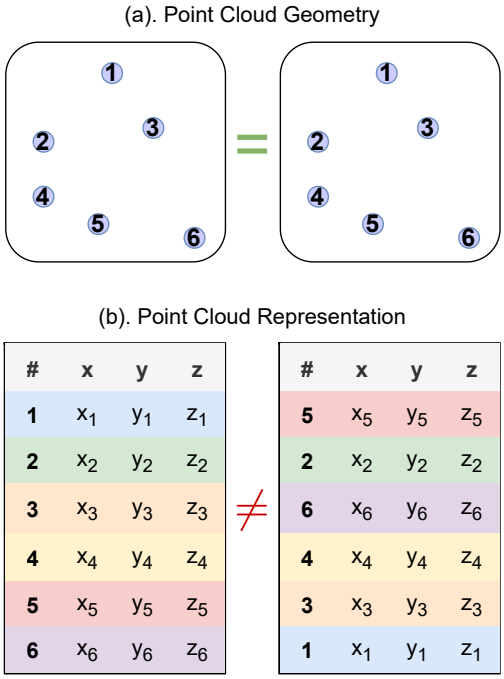


Figure 6.8: The same point distribution in the geometry in (a) can share multiple point cloud representations. Two representations are shown in (b) as an example.

6.4.1.4 Bi-LSTM Block

Classifying individual frames containing only the body-shaped point cloud is less efficient. Instead, employing a data sample that contains multiple adjacent frames in a time sequence offers a more comprehensive representation of the human target, incorporating not only body shape information but movement characteristics (e.g., gait) [49]. To enable the system to extract features from both these aspects, the Bi-LSTM structure which is well-suited for sequential data feature extraction is integrated.

To add the Bi-LSTM Block after the three modules, an additional layer called the TimeDistributed layer, which wraps the previous three modules, is required as shown in Figure 6.7. This TimeDistributed layer allows the layer being wrapped (e.g., the CNN layers) to apply the same operation to each time step of the sequence input and retain the time dimension T. The extracted features are then flattened and provided to the Bi-LSTM model for further processing.

6.4.1.5 Dense Block

Typically positioned at the end of the network structure, dense layers play a role in flattening and reorganizing the previously extracted features. To mitigate overfitting, dropout layers with a dropout rate of 0.3 are interspersed between each dense layer. The final dense layer is followed by a softmax activation function, employed to output the classification categories K (i.e., $K = 12$).

6.4.2 Network Training

The system is built using Python (version 3.8.6) and TensorFlow (version 2.6.5), with key dependencies such as Keras, NumPy, and h5py, on a desktop PC equipped with an Intel(R) Core (TM) i7-10850H CPU @ 2.70GHz and an NVIDIA RTX 2080 GPU with 8GB dedicated memory. CUDA is also required to enable GPU acceleration for the network training process. To access faster training and larger dedicated GPU memory, the network training process is transferred to Google Colab [203], which utilizes an A100 GPU with 40GB of dedicated memory online.

The hyperparameters such as the number of hidden units and kernel size for various network layers (CNN, Bi-LSTM, dense, and dropout) are annotated in Figure 6.7. The loss function is sparse categorical cross-entropy, and the optimizer is Adam. A training batch size of 128 is determined due to limited memory and the network is trained for 200 epochs with a learning rate of 0.0002. For the code implementation, please refer to Appendix A.

6.4.3 Prediction with Window Sliding

The network predictions may exhibit insufficient confidence in assigning the correct label occasionally, resulting in mislabelled segments within the prediction sequence. To further enhance the system accuracy, a window sliding approach is employed on the prediction sequence, opting for a window length of 12 predictions (approximately 3 seconds). This method selects the majority category within the window to reduce the impact of minority wrong predictions, which are highlighted in red in Figure 6.9.

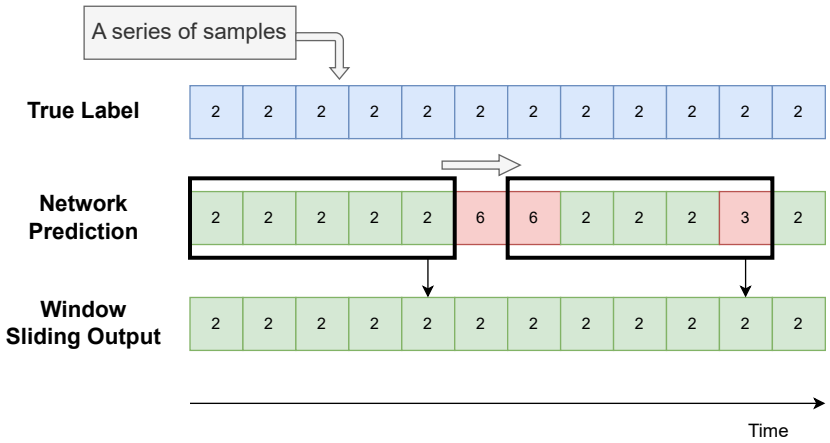


Figure 6.9: The comparison is depicted between the network prediction and the result obtained after applying a window sliding operation, with a window length of 5 predictions as an example.

Given that human identities are consistent when an individual is present in the room, the window sliding approach proves effective. Leveraging the previously developed tracking system in Chapter 5, the network results can be accumulated over time. This system enables the accurate

classification of human identity once a person is detected and tracking is initiated. When the person exits the room, the system deactivates tracking and clears the stacked sample results in the sliding window to prepare for the next tracking instance.

6.5 System Evaluation

Approximately 20 minutes of data is generated for each human subject, resulting in a total of 240 minutes (equivalent to 288,000 frames). The system accuracy in classifying the identities of 12 individuals is evaluated, considering scenarios where only one person is present at any given time.

6.5.1 Identification Performance

By employing the MMIDNet network, the performance of the system on the test set achieves an overall accuracy of 92.4% with the average F1 Score of 92.0%. The accompanying confusion matrix in Figure 6.10 demonstrates a robust performance in distinguishing the identities of the 12 individuals. Most human subjects exhibit high classification accuracy, exceeding 90%.

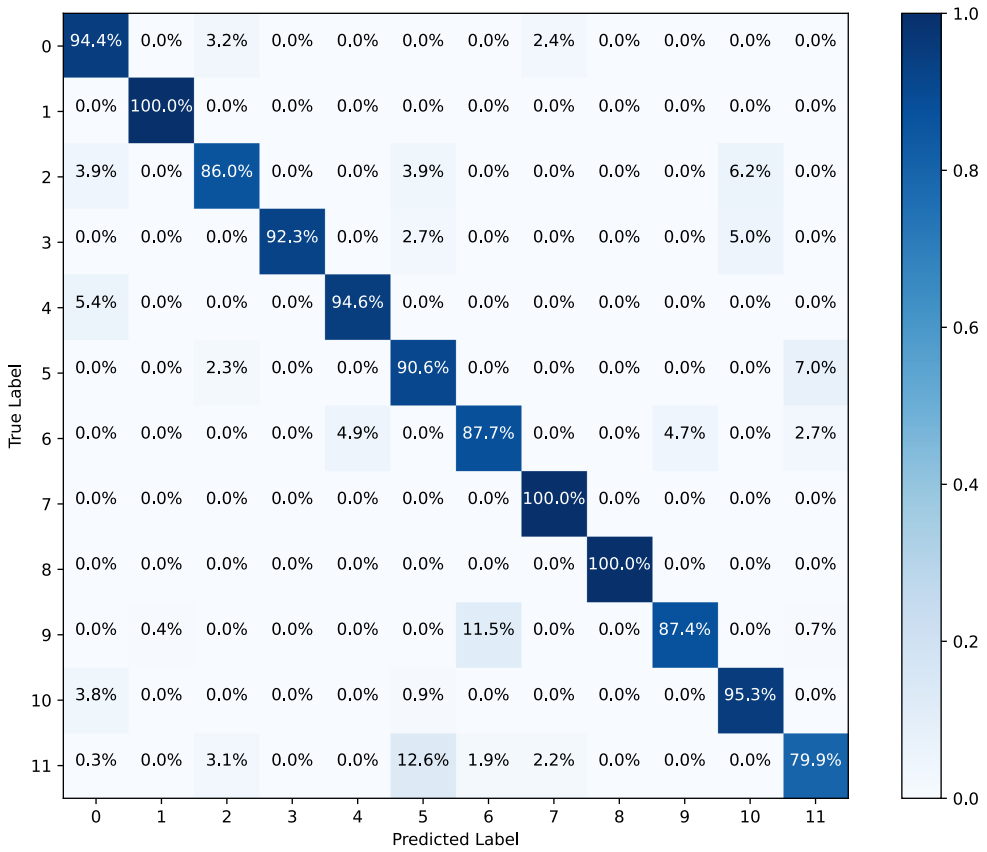


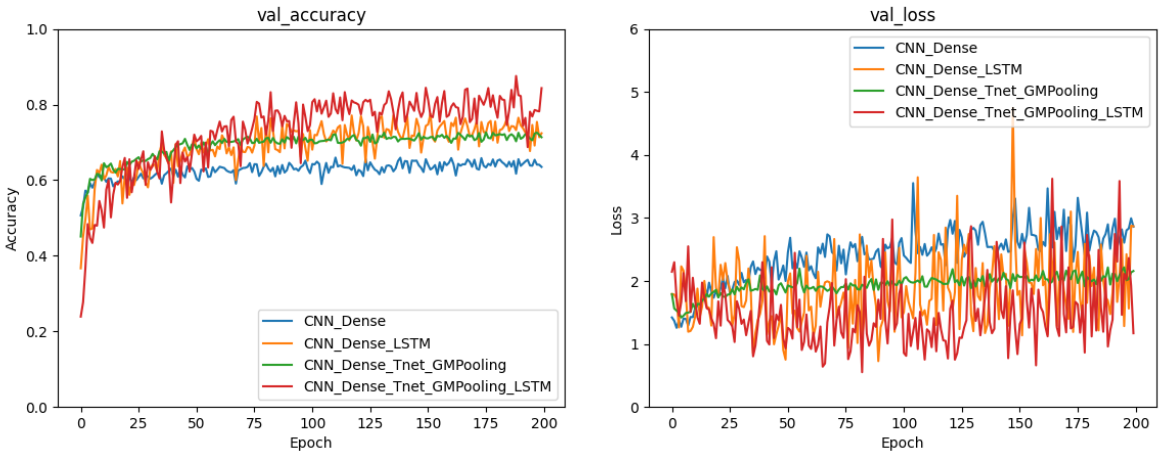
Figure 6.10: The confusion matrix for the classification prediction results of 12 people's identities.

One subject (category label 11) falls below 80% as shown in Figure 6.10. The confusion matrix displays a 12.6% chance of misclassifying label 11 as label 5, suggesting the system struggles to distinguish between these labels due to similar body shapes, similar walking gaits, and insufficient data. On the other hand, label 5 also has a relatively higher chance (7.0%) of being misclassified as label 11. A similar issue occurs between label 6 and label 9, slightly lowering the overall accuracy. To address this and improve accuracy, more data needs to be collected and fed into the network in the future.

Meanwhile, three categories (labels 1, 7, and 8) achieve 100% accuracy in the confusion matrix, indicating the system is highly confident in classifying them. One of these subjects is particularly tall at 185 cm, causing their point cloud to expand more along the z-axis compared to other subjects. Another subject is relatively heavy at 80 kg, resulting in a wider point cloud in both the x and y axes. These distinctive point cloud features make them easier to distinguish.

6.5.2 Evaluation of Network Structure

To investigate the impact of the network structure on human identification performance, various network configurations are evaluated, and the accuracy is measured when specific components are removed. This approach allows for a detailed analysis of the contribution of each component to the overall system performance. Figure 6.11 demonstrates the trajectory of accuracy and loss for the different network configurations on the validation set throughout the training process spanning 200 epochs. Obviously, the basic CNN + Dense structure maintains at 60% accuracy, with its loss rising after 25 epochs, indicating a potential overfitting issue. Based on this, introducing T-Net and Global Max Pooling components adopted from PointNet [128] enhances accuracy to



(a). Validation Accuracy **(b). Validation Loss**

Figure 6.11: The accuracy (a) and loss (b) for different network configurations on the validation set over epochs.

approximately 70%, while applying a Bi-LSTM component yields a comparable outcome with slight variations. The red line in Figure 6.11 represents the combination of all components, resulting in an accuracy of around 80%, peaking at 87% at the epoch of 188.

The weight checkpoints are saved at each epoch during network training. After the training is complete, the epoch checkpoints with the best weights are reloaded for each configuration and used to perform inferences on the test set. Table 6.1 shows the performances of these configurations on the test set. The basic configuration of CNN and Dense layers can only reach an overall accuracy of 61.1%. Incorporating T-Net and Global Max Pooling, enhances accuracy by approximately 10%. The addition of Bi-LSTM further improves accuracy by about 16%. Ultimately, applying the window sliding strategy after the network results in the full system achieving an accuracy of 92.4% for 12 identities.

Table 6.1: Different network structure accuracy comparison on the test set.

Network structure	Accuracy
CNN + Dense	61.1%
CNN + Dense + Tnet + Global Max Pooling	70.5%
CNN + Dense + Bi-LSTM	73.0%
CNN + Dense + Tnet + Global Max Pooling + Bi-LSTM	86.3%
Full System	92.4%

6.5.3 Evaluation of Observation Duration

Common intuition suggests that the longer the sensor monitors, the greater the likelihood of accurately identifying them. To evaluate this, the impact of extended monitoring on system performance is examined in the context of identifying 12 individuals. To be more specific, as outlined in Section 6.4.3, a 3-second observation duration is used to generate predictions, yielding an overall accuracy of 92.4%, as shown in Figure 6.12. When the observation duration is extended to 7 seconds, the system achieves a significantly higher accuracy of over 98% on the test set.

Figure 6.12 illustrates the accuracy of the system across different observation durations, ranging from 3 to 7 seconds. The blue bars represent overall accuracy values, while the black boundaries indicate the range between the highest and lowest accuracy values for different labels. As the observation time increases, there is a consistent rise in system accuracy and a narrowing

in the variability between maximum and minimum accuracy values, demonstrating enhanced robustness and reliability with longer observation periods. Although a longer observation duration slows down the system, this delay only impacts the first prediction when a human subject first appears in the scene. It does not necessitate another full 7-second wait for subsequent predictions.

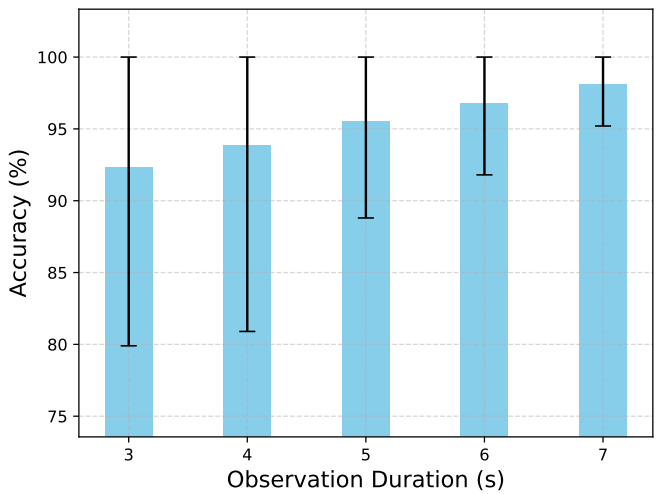


Figure 6.12: The system accuracy across various lengths of observation time, ranging from 3 to 7 seconds.

6.5.4 Evaluation of Number of People

Additionally, an assessment is performed to classify identities within various group sizes, as depicted in Figure 6.13. The system demonstrates strong performance, particularly with smaller groups ranging from 4 to 8 individuals, consistently achieving an accuracy of approximately 97%. The highest overall accuracy surpasses 98% when the group size is 4, with a minimum accuracy of 95%, indicating exceptional reliability in scenarios with fewer occupants.

This contrasts sharply with the results from Zeng et al. [2], who achieved only 80% accuracy when identifying 6 individuals using WiFi CSI. The significant improvement in accuracy using mmWave radar highlights the system's robustness and precision, especially when faced with smaller groups, where it is able to effectively capture and distinguish the unique characteristics of each individual. This outcome underscores the system's applicability in smart home and office environments, where the number of individuals present is typically limited. Moreover, even as the group size increases to 12 people, the system's performance can still remain efficient.

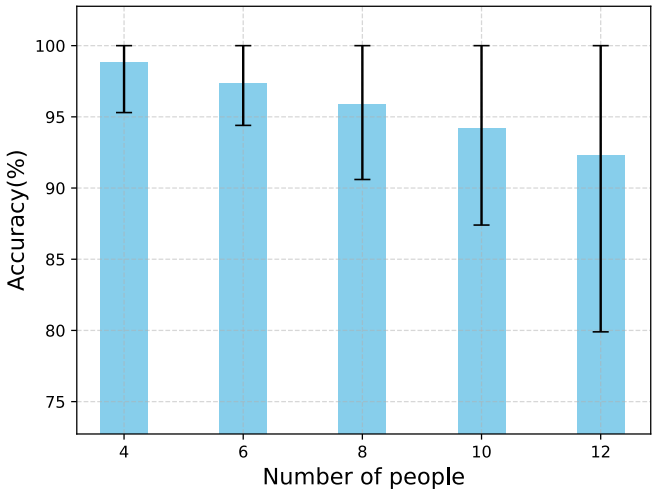


Figure 6.13: The system accuracy across various scenarios involving the classification of different numbers of people, ranging from 4 to 12.

6.5.5 System Comparison

Table 6.2 presents a comparative analysis of human identification systems. Sokolova and Konushin [102] introduced a solution involving cameras and neural networks for human identity classification, achieving outstanding performance exceeding 98% accuracy on the TUM-GAID dataset. Meanwhile, Koide et al. [104] employed cameras with depth information and integrated Bayesian inference methods, achieving a 98.4% accuracy for 6 subjects. However, both approaches in [102] and [104] face severe privacy concerns due to intrusive camera usage.

Table 6.2: Human identification system comparison.

	Camera [102]	RGB-D Camera [104]	Floor sensor [117]	WiFi WiWho [2]	MmWave Radar mID [49]	MmWave Radar MMIDNet (This system)
ID classification	**Acc. > 98% on TUM-GAID**	87.9% on 6 subjects, 98.4% with Bayesian	Acc. 80% on 15 subjects	92% on 2 subjects, 80% on 6 subjects	**Acc. 89% on 12 subjects**	**Acc. 92.4% on 12 subjects**
Multiple people	Not consider	**Yes**	Not consider	No	**Yes**	**Yes**
NN usage	**Yes**	**Yes**	No	No	**Yes**	**Yes**
Inference speed	-	> 4.2s	**22Hz**	**0.2s**	**> 2s**	**3s Acc. 92.4%, 7s Acc. > 98%**
Input data size	Large (images)	Moderate (image segments)	**Small**	Moderate (CSI)	Large (3D voxelized matrix)	**Small (Sparse point cloud)**
Privacy concerns	Severe	Severe	**Negligible**	**Low**	**Low**	**Low**
Deployment	**Easy**	Moderate	Difficult	**Easy**	Moderate	**Easy & extendable**

To address these privacy issues, Middleton et al. [117] developed a floor sensor matrix to record human footprints, a feature considered less invasive. Without relying on neural networks,

they achieved 80% accuracy for 15 subjects, operating at 22Hz. However, the deployment of this floor sensor is difficult and troublesome. Another non-intrusive approach is proposed by Zeng et al. [2], who achieved 92% accuracy for 2 subjects and 80% for 6 subjects on human identification by analyzing WiFi CSI. Although deployment is straightforward, it requires human movement between TX and RX which are separate devices. Recently, mmWave radar has garnered attention for non-intrusive human identification by Zhao et al. [49]. They proposed a system called mID that is designed for human tracking and identification. By voxelizing the point cloud provided by mmWave radar and feeding it into the network, mID achieved 89% accuracy for 12 subjects. However, point cloud voxelization leads to a large input dataset which makes it difficult for network training.

In contrast, the MMIDNet proposed in this work is a neural network structure optimized for mmWave radar applications. It utilizes sparse point cloud data, characterized by its small size, as input to expedite the network training process to learn useful information. With a duration of 3 seconds, this system achieves an accuracy of 92.4% for 12 subjects. Extending the observation time to 7 seconds yields a higher accuracy of 98%. Furthermore, the MMIDNet can discern multiple targets within the same scene by implementing clustering algorithms to distinguish between human targets. Integration with the previous work in Chapter 5 allows for extensibility by incorporating additional radars into the system.

6.6 Conclusion

This chapter introduces MMIDNet, a deep neural network structure crafted for mmWave radars in human identification tasks. It begins with a detailed explanation of the data collection process and the formation of datasets for training, validation, and testing. Subsequently, this chapter demonstrates each network component within the MMIDNet architecture and expounds on their respective contributions to the methodology. The proposed method achieves an impressive overall accuracy of 92.4% in classifying 12 human identities. With an extended observation duration of 7 seconds, the accuracy exceeds 98%. Additionally, the chapter explores the effects of different network configurations and their impact on identifying varying numbers of individuals. Furthermore, a comparative analysis with other state-of-the-art approaches is provided in aspects of performance, privacy and deployment. Although additional experiments across various scenarios are necessary for comprehensive human identification practices in the future, this study lays the groundwork for the feasibility of utilizing mmWave radars and deep learning methodologies in smart home and office settings. This work contributes significantly to the preservation of privacy in these environments.

Future work involves expanding the dataset to improve the accuracy and cover more complex situations. Meanwhile, classifying multiple human subjects within the same room at the same time can be achieved by integrating with the previous real-time system in Chapter 5.